



King Saud University
College of Business Administration
Management Information Systems Department



Project SmartFridge
Project Management in IT - MIS 460

Dr. Huda Alhomoud

Section: 66983, 8:00pm

Group Name	ID number
Leader- Jumanh AlGhamdi	444201420
Aljazi Alamro	444201329
Rahaf Alharbi	433200770
Ghada Alshehri	444202103
Deem Althawab	444201627

Table of content

Introduction.....	4
Mission.....	4
Vision	4
Preliminary Research.....	4
SmartFridge - Benchmark	5
Stakeholders register and Management strategy.....	6
Stakeholder Register	6
Stakeholder Management Strategy	6
Team Contract:.....	8
Code of Conduct:.....	8
Participation:	9
Communication:	9
Collaboration:	9
Problem Solving:	9
Compliance with Policies and Regulations:	10
Meeting Guidelines:.....	10
Project Charter	11
Business Case.....	12
Requirements Traceability Matrix for SmartFridge.....	16
Scope Statement.....	17
Cost Estimate	19
Work Breakdown Structure:.....	20
Milestone List :	22
Gantt chart.....	24
Network diagram	25
Business Plan	26
Executive Summary.....	26
Company Description	26
Products and Services	26
Marketing Plan	27
Operational Plan.....	27
Revenue Model	28

Business Model.....	28
probability/impact matrix.....	29
List of Prioritized Risks:.....	29
Updated List of Prioritized Risks.....	30
Risk Register.....	30
Lessons Learned Report:.....	31
Best Practices:.....	33
conclusion	33
References:.....	34

Introduction

SmartFridge is a technology startup that develops an AI-powered smart refrigerator system designed to help households and restaurants manage food efficiently. The system uses sensors and artificial intelligence to track food expiration dates, send alerts before products expire, and suggest recipes based on available ingredients. By integrating smart technology into everyday kitchen use, SmartFridge aims to simplify food management and reduce unnecessary waste.

Mission

At SmartFridge, our mission is to help households and restaurants reduce food waste by providing an intelligent and easy-to-use system that monitors food freshness, sends timely notifications, and offers practical recommendations for better food usage.

Vision

Our vision at SmartFridge is to become a leading smart kitchen solution provider in Saudi Arabia and globally, creating a future where food waste is minimized through advanced, connected, and intelligent home technologies.

Preliminary Research

The smart home and kitchen appliance market in Saudi Arabia is experiencing rapid growth due to increased demand for convenience, automation, and modern lifestyles. Government initiatives under Vision 2030 have accelerated digital transformation, encouraging the adoption of smart technologies across households. As a result, consumers are increasingly interested in connected kitchen solutions that improve efficiency and reduce manual effort.

The kitchen appliances market in Saudi Arabia was valued at approximately USD 523 million in 2024 and is expected to grow steadily in the coming years. Within this market, smart appliances, particularly smart refrigerators, are gaining attention due to their ability to connect with mobile applications and provide remote control features. However, despite this growth, many of these products still lack advanced capabilities related to food management and waste reduction.

Major companies such as Samsung and LG dominate the smart appliance sector by offering smart refrigerators with basic connectivity features. These refrigerators allow users to monitor temperature, access internal cameras, and manage settings remotely. However, their focus is primarily on convenience and connectivity rather than intelligent food tracking or personalized recommendations.

Food waste remains a significant challenge in Saudi Arabia, with a large amount of food wasted daily at the household level. This creates both economic and environmental concerns, highlighting the need for innovative solutions that help users better manage their food consumption.

SmartFridge - Benchmark

Category	SmartFridge	Samsung	LG
Type	AI-powered smart refrigerator system	Smart refrigerator (IoT-based)	Smart refrigerator (IoT-based)
Core Features	- Tracks food expiration dates - Sends alerts before products expire - Suggests recipes based on available ingredients - Integrates with smart refrigerators - Temperature control	-Remote Monitoring -Temperature control -Internal cameras	- Remote Monitoring -Temperature control -Internal cameras
Technology Used	AI + Sensors	IoT	IoT
Food Management	Advanced tracking + recommendations	Basic monitoring	Basic monitoring
Limitations	New in market	No AI tracking	No AI tracking
Unique Value	Fully automated food management system	Focus on appliance control	Focus on connectivity

Stakeholders register and Management strategy

Stakeholder Register

Date:3/4/2026

Prepared by: Rahaf Al-Harbi

Name	Position	Internal/External	Project Role	Contact Information
Rahaf Al-Harbi	PMO director	Internal	Project Manager	rahaf @SmartFridge.com
Jumanh Al-Ghamdi	Business Analyst	Internal	Team Member	jumanh @SmartFridge.com
Aljazi Al-Amro	AI developer	Internal	Team Member	aljazi @SmartFridge.com
Deem Al-Thawab	Senior Financial	Internal	Coordinator	deem @SmartFridge.com
Ghada Al-Shehri	Electrical engineer	Internal	Advisor	ghada @SmartFridge.com
Nada Al-Harhi	Investor	External	Sponsor	nada@gmail.com
Al Rajhi Bank	Financial Institution	External	Financial Backer	care@alrajhibank.com.sa
Samsung	Electronics Company	External	Global competitor	info@samsung.com
customer	End user	External	End user	-----

Stakeholder Management Strategy

Date:3/4/2026

Prepared by: Rahaf Al-Harbi

Rahaf Al-Harbi	High	High	-She is highly responsible and detailoriented ,and has a strong interest in the project . -Give her decision-making authority .
----------------	------	------	--

Jumanh Al-Ghamdi	High	Medium	-she has an interest in this project. She needs this project to be a milestone in her career . -Explain to her how much this project will help her in her career .
Aljazi Al-Amro	High	Medium	-She has experience in AI and IOT. She has another project she is working on . -Organizing her tasks at times that suit her schedule .
Deem Al-Thawab	High	Medium	-She is very dedicated to her work. She is very precise in her calculations. It's her first time working, so she's excited to work on this project. - Explain to her the importance of her presence in this job and have her perform the duties of coordinator and financier at the same time.
Ghada Al-Shehri	High	Medium	-She is interested in practical and field related work . -Assign her clear technical tasks and monitor her performance regularly .
Nada Al-Harthi	High	High	-She likes to be aware of all the details . -She is interested in making money . -Conduct weekly meetings and provide regular progress reports to manage her closely .
Al Rajhi Bank	Medium	High	-The bank is interested in stable financial returns and low risk . -Provide financial reports and ensure timely payments to keep them satisfied .
Samsung	Low	High	-A global competitor with strong influence in the market but low direct interest in the project . -Monitor their products and analyze market trends regularly .

customer	High	Medium	-They care about the product having a reasonable price . -They care about the quality and safety of the product . -Prove them with industry standards that it is safe to use . -Prove them through experience that it is of high quality .
-----------------	------	--------	---

Team Contract:

Project Team Members Names and Sign-off:

Name	Signature
Nada Al-Harhi	
Rahaf Al-Harbi	
Jumanh Al-Ghamdi	
Aljazi Al-Amro	
Deem Al-Thawab	
Ghada Al-Shehri	

Code of Conduct:

- Work together as a team to fairly divide the project's workload.
- Encourage teamwork among participants to guarantee timely task completion while aiming to fulfill deadlines.
- Throughout the project, offer encouragement and support to other team members.
- Remain committed to the project's objectives and collaborate with others to make them a reality.
- Maintain transparent and efficient communication by providing all team members with up-to-date project information.
- Encourage a culture of information sharing and teamwork by sharing our unique skills, knowledge, and ideas with the entire team.

Participation:

Our commitment includes:

- Ensuring our presence and readiness for all project-related meetings and engagements.
- Engaging actively in discussions to explore and analyze the various stages of the project.
- Providing equal opportunities for every team member to actively participate and contribute.
- Embracing an open mindset towards all project activities and decisions.
- Demonstrating respect towards all participants in meetings, valuing their perspectives and opinions.

Communication:

Our commitment involves:

- Utilizing online communication tools (WhatsApp group, Microsoft Teams) as our primary means of interaction.
- Conducting regular weekly team meetings every Saturday at 6:00 PM to discuss project progress.
- Ensuring that our written and verbal communication is clear, concise, and respectful.
- Practicing active listening and prioritizing understanding others' perspectives before formulating responses.
- Responding to all messages and requests within 24 hours on weekdays.

Collaboration:

Our commitment involves:

- Cultivating an environment of collaboration and cooperation throughout the project lifecycle.
- Promoting the exchange of ideas, knowledge, and resources among all team members.
- Prioritizing teamwork and actively contributing to each other's achievements and growth.
- Using Microsoft Teams and Google Drive for document collaboration and version control.

Problem Solving:

Our approach includes:

- Encouraging each team member's active contribution towards identifying and solving challenges.

- Collaboratively generating ideas and solutions to address obstacles and optimize outcomes.
- Ensuring the involvement and consideration of relevant stakeholders in the problem solving process.
- Applying structured root cause analysis and pre-mortem analysis to identify potential risks proactively.

Compliance with Policies and Regulations:

Our commitment involves:

- Abiding by all relevant laws, regulations, and policies set forth by King Saud University and applicable authorities.
- Ensuring strict adherence to course guidelines and academic integrity standards (MIS 460).
- Fulfilling all contractual obligations and submission requirements on time.

Meeting Guidelines:

Our guidelines include:

- The Project Manager (Rahaf Al-Harbi) will schedule and chair all meetings.
- Regular meetings will be conducted via Microsoft Teams; quick updates via WhatsApp.
- Each meeting will have a clear agenda distributed at least 24 hours in advance.
- Meeting minutes will be documented and shared within 24 hours after each meeting.
- Absence must be communicated to the PM at least 12 hours prior to the meeting.

By recognizing and following this code of conduct, every team member actively contributes to fostering a positive and productive work environment for the SmartFridge project.

Project Charter

Project Title :SmartFridge – AI-Powered Food Expiration Tracking

Project Start Date: April

2026 Projected Finish Date :April2027

Budget information :The project has an initial budget of SAR 40,000. However, the cost is projected to exceed the budget by SAR 100,000, bringing the total estimated cost to SAR 140,000(Cost = 40 .)100,000 + 000,The additional funding will be covered through investor Nada Al-Harhi and financial backing from Al Rajhi Bank. The majority of costs will cover AI development, sensor hardware integration, software engineering, and system testing. A detailed budget breakdown will be prepared by the project Manager (Rahaf Al-Harbi) .

Project Manager: Rahaf Al-Harbi, rahaf@SmartFridge.com

Project Objectives :Develop an AI-powered SmartFridge system designed to help households and restaurants in Saudi Arabia track food expiration dates using IoT sensors and artificial intelligence. The project is planned to be completed within 12 months. The system will send automated notifications to users before food items expire, suggest personalized recipes based on available ingredients, and integrate seamlessly with smart refrigerators to manage food inventory efficiently. The solution targets the growing Saudi smart home market and aligns with Vision 2030 digital transformation goals, aiming to reduce food waste — currently estimated at 1.3 kg per person per day — and lower costs for households and businesses .

Main Project Success Criterion: The SmartFridge system should achieve a measurable reduction in household food waste, gain adoption by at least 50 users within six months of launch, and generate sufficient revenue to cover operational costs within one year of project completion.

Approach :

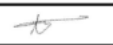
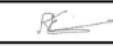
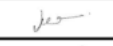
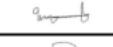
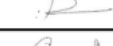
- Conduct market research and user surveys to identify critical features needed for the Saudi smart kitchen segment .
- Review existing smart appliance solutions and identify gaps in food expiration tracking and inventory management .
- Develop AI algorithms for real-time food expiration tracking and ingredient-based recipe suggestions .
- Research and integrate IoT sensor technology with smart refrigerator hardware for automated inventory monitoring .
- Build the system using an iterative development approach across 12 months, continuously soliciting user feedback to refine features .

- Measure the value of the system through reduction in food waste and revenue generation, both during the project and one year after completion .

ROLES AND RESPONSIBILITIES (PARTIAL LIST)

Name	Position	Internal/External	Project Role	Contact Information
Rahaf Al-Harbi	PMO director	Internal	Project Manager	rahaf@SmartFridge.com
Jumanh Al-Ghamdi	Business Analyst	Internal	Team Member	jumanh@SmartFridge.com
Aljazi Al-Amro	AI developer	Internal	Team Member	aljazi@SmartFridge.com
Deem Al-Thawab	Senior Financial	Internal	Coordinator	deem@SmartFridge.com
Ghada Al-Shehri	Electrical engineer	Internal	Advisor	ghada@SmartFridge.com
Nada Al-Harthi	Investor	External	Sponsor	nada@gmail.com
Al Rajhi Bank	Financial Institution	External	Financial Backer	care@alrajhibank.com.sa
Samsung	Electronics Company	External	Global competitor	info@samsung.com
customer	End user	External	End user	-----

:Sign-off

Name	Signature
Nada Al-Harthi	
Rahaf Al-Harbi	
Jumanh Al-Ghamdi	
Aljazi Al-Amro	
Deem Al-Thawab	
Ghada Al-Shehri	

Business Case

Date: April 6, 2026

1.0 Introduction/ Background

To improve the way households and restaurants manage food, a team of professionals in Management Information Systems, artificial intelligence, and engineering came together to develop SmartFridge. The team recognized the growing issue of food waste and the lack of intelligent systems that help users monitor food expiration and manage their inventory efficiently.

SmartFridge aims to provide a smart solution that combines artificial intelligence and Internet of Things (IoT) technologies to track food expiration dates, send alerts before items expire, and suggest recipes based on available ingredients. The system is designed to simplify daily food management and reduce unnecessary waste.

2.0 Business Objective

To position SmartFridge as a leading smart kitchen solution, the project aims to reduce food waste and improve efficiency in food management. The system will help users make better decisions by providing timely notifications and intelligent recommendations.

The project also aims to support digital transformation in Saudi Arabia by introducing innovative smart home technology. In addition, it seeks to generate sustainable revenue through product sales, subscriptions, and partnerships with appliance companies.

3.0 Current Situation and Problem/Opportunity Statement

Managing food efficiently is a challenge for many households and restaurants. Users often rely on manual methods to track food expiration dates, which leads to forgotten items and increased food waste. In addition, over-purchasing food contributes to financial loss and environmental impact.

Although smart refrigerators exist in the market, such as those offered by Samsung and LG, they mainly focus on connectivity and basic features rather than intelligent food tracking. Other solutions focus on redistributing food rather than preventing waste at the source.

SmartFridge presents a strong opportunity to address this issue by providing a comprehensive system that combines AI and IoT to deliver real-time tracking, automated alerts, and personalized recommendations.

4.0 Critical Assumption and Constraints

Assumptions:

1. Users are willing to adopt smart technologies in their daily lives.
2. The system will provide accurate food tracking and recommendations.
3. The smart interface on the fridge will be user-friendly and easy to use.
4. Users will respond to notifications and suggestions. **Constraints:**
 1. High cost of integrating hardware and software components.
 2. Dependence on internet connectivity.
 3. Possible resistance from users unfamiliar with smart technology.
 4. Technical challenges in system integration.

5.0 Analysis of Option and Recommendation

1. Continue using traditional methods of food management without implementing a new system.
2. Develop a basic smart fridge interface (built-in screen) without integrating advanced hardware or AI features.
3. Design and implement a complete system using artificial intelligence and IoT technologies.

Based on discussions with project stakeholders, option 3 has been selected as it provides a comprehensive and innovative solution that effectively addresses the problem.

6.0 Preliminary Project Requirements

1. Develop an AI system for tracking food expiration dates.
2. Integrate IoT sensors with smart refrigerators.
3. Build a smart fridge interface (built-in screen) for user interaction.
4. Implement an automated notification system.
5. Provide recipe recommendations based on available ingredients.
6. Ensure system security and data protection.
7. Establish effective project management practices.

9.0 Potential Risks

1. Failure to complete the project within the planned time.
2. Budget exceeding the estimated cost.
3. Technical issues in system development and integration.
4. Low user adoption of the system.
5. Data security and privacy risks.
6. Competition from established companies.

10.0 Exhibits

Exhibit A: Financial Analysis

Discount rate	8.00%				
Assume the project is completed in Year 0					
	0	1	2	3	Total
Costs	140,000	40,000	40,000	40,000	
Discount factor	1.00	0.93	0.86	0.79	
Discounted costs	140,000	37,200	34,400	31,600	243,200
Benefits	0	200,000	200,000	200,000	
Discount factor	1.00	0.93	0.86	0.79	
Discounted benefits	0	186,000	172,000	158,000	516,000
Discounted benefits - costs	(140,000)	148,800	137,600	126,400	272,800 ← NPV
Cumulative benefits - costs	(140,000)	8,800	146,400	272,800	
ROI	112%				
	Payback in Year 1				
Assumptions					
Cost	#hours				
PM (500 hours, 30 SAR/hour)	15,000				
Technical Staff (AI, IoT, Developers)	70,000				
Equipment (Smart fridge components)	40,000				
IoT sensors, software, and system integration	15,000				
Total project costs	140,000				
Benefits					
Revenue from SmartFridge system sales and services	200,000				
Total annual project benefits	200,000				

7.0 Budget Estimate and Financial Analysis

The total cost estimate for the SmartFridge project is SAR 140,000. This budget covers all major project components, including project management, development of AI algorithms, technical staff efforts, integration of IoT sensors, software development, system testing, and the required equipment for smart fridge components.

The projected benefits of the project are based on revenue generated from SmartFridge product sales. It is estimated that the system will generate approximately SAR 200,000 per year.

Overall, the total project benefits amount to approximately SAR 200,000 annually. The financial analysis shows that the estimated Net Present Value (NPV) is SAR 272,800, and the Return on Investment (ROI) is 112%. The payback period is within the first year, as expected.

8.0 Schedule Estimate

The project is expected to be completed within 12 months, with some flexibility in the schedule. It is also assumed that the system will have a useful life of at least three years.

Requirements Traceability Matrix for SmartFridge

Prepared by: Rahaf Alharbi

Date: April 6, 2026

Requirement No.	Name	Category	Source	Status
1	AI Food Expiration Tracking System	Functional	Project charter and business case	Feature is currently under development by the AI developer.
2	IoT Sensor Integration with Smart Fridge	Technical	Project charter and technical specifications	Sensor integration is in progress; hardware components have been procured.
3	Automated Expiry Alert Notifications	Functional	Stakeholder meeting and scope statement	Notification system is under development and pending integration testing.
4	Recipe Recommendation Feature	Functional	Business case and scope statement	Feature has been developed and tested successfully.
5	Smart Built-in Fridge Screen Interface	Technical	Project charter and product design documents	Screen interface design is complete; hardware installation is in progress.
6	Mobile App for Remote Access	Technical	Scope statement and stakeholder requirements	Mobile application is under active development for iOS and Android platforms.
7	Monthly Subscription Feature	Functional	Business plan and financial projections	Subscription model has been defined; payment integration has not yet started.

8	Online Payment Gateway (Visa/Mada/Apple Pay)	Technical	Business plan and financial requirements	Payment gateway selection is complete; technical integration has not yet begun.
9	Data Security and Privacy Protocols	Technical	Project charter and data protection regulations	Security protocols have been defined and implemented successfully.

Scope Statement

Project Title: SmartFridge – AI-Powered Food Expiration Tracking System	
Date: April 6, 2026	Prepared by: Rahaf Alharbi
Project Justification: Managing food efficiently is a daily challenge for households and restaurants in Saudi Arabia. Users currently rely on manual methods to track expiration dates, which frequently results in forgotten items, avoidable food waste, and financial loss. Although smart refrigerators from companies such as Samsung and LG exist in the market, they focus primarily on basic connectivity features such as temperature control and remote monitoring, rather than intelligent food tracking or personalized recommendations. SmartFridge addresses this gap by developing a fully integrated AI and IoT system that tracks food expiration in real time, sends automated alerts before items expire, and suggests recipes based on available ingredients — reducing waste, saving money, and supporting Saudi Arabia's Vision 2030 digital transformation goals.	
Product Characteristics and Requirements: Food Expiration Tracking: The system must be capable of tracking the expiration dates of all food items stored in the refrigerator in real time. A minimum of 5 IoT temperature and freshness sensors will be installed inside the fridge to monitor stored items. The AI algorithm must achieve a tracking accuracy rate of at least 90%. All expiration data must be updated automatically and displayed on the built-in screen without any manual input from the user Automated Alert Notifications: The system must send automated alerts to the user at least 48 hours before any food item is due to expire. Alerts must be categorized by urgency: a yellow alert at 48 hours before expiry and a red alert on the day of expiry. Notifications must be displayed on the built-in screen. The user must be able to customize alert preferences, including the notification timing and the types of food items they wish to be notified about.	

3. **Recipe Recommendation Engine:** The AI-driven recipe recommendation feature must analyze the available ingredients currently stored in the fridge and suggest at least 3 suitable recipes per day based on those ingredients. Recipes must be prioritized according to which ingredients are closest to their expiration date, encouraging the user to use those items first. The feature must allow users to filter recommendations based on dietary preferences such as vegetarian, gluten-free, or low-calorie options. Recipe suggestions must be accessible from the built-in fridge screen.
4. **Smart Built-in Fridge Screen Interface:** The system must include a 10-inch touchscreen display built into the front panel of the refrigerator. The screen must display the current food inventory, expiration alerts, recipe suggestions. The interface must support Arabic and English languages. The screen must be responsive and load any page within 2 seconds. All interface interactions must be intuitive and require no technical knowledge from the user.
5. **Data Security and Privacy:** All user data, including food inventory records, dietary preferences, and payment information, must be encrypted using AES-256 encryption both in transit and at rest. The system must comply with applicable Saudi data protection regulations. User data must not be shared with any third party without explicit user consent. A data backup must be performed automatically every 24 hours to prevent data loss in the event of a system failure.

Summary of Project Deliverables

Project management-related deliverables: business case, project charter, team contract, scope statement, WBS, schedule, cost baseline, status reports, final project presentation, final project report, lessons-learned report, and any other documents required to manage the project.

Product-related deliverables: research reports, design documents, AI and IoT software code, hardware (IoT sensors and smart fridge screen), testing and quality assurance reports, and fully integrated SmartFridge prototype.

1. AI food expiration tracking and automated notification system
2. IoT sensor hardware and 10-inch smart fridge built-in touchscreen
3. Fully integrated SmartFridge prototype ready for demo and user testing

Project Success Criteria:

1. Successfully develop and deploy the AI food expiration tracking system with full IoT sensor integration within the 12-month project timeline (April 2026 – April 2027).
2. Achieve at least 50 active paying **customers** within 6 months of the official product launch.
3. Keep total project costs within the approved SAR 140,000 budget secured through the sponsor and Al Rajhi Bank, with the possibility of seeking additional funding if required.

4. Attain a measurable reduction in household food waste among active users, validated through post-launch surveys conducted 3 months after launch.
5. Receive an average user satisfaction rating of at least 4 out of 5 based on built-in screen evaluations and user feedback surveys.
6. Obtain formal written approval and sign-off from the project sponsor (Nada Al-Harthi) upon project completion.

Cost Estimate

SmartFridge Project Cost Estimate

Prepared by: Rakaf Al-Harbi

Date: April 2024

Note: Change the WBS items and other entries to meet your project needs. This data is from Figure 7-1 of Schwab's text Information Technology Project Management, Fourth Edition. Also make sure the formulas work properly based on the data you enter.

WBS Items	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	WBS Level 1 Totals	% of Total
1. Initiating				\$2,500	1%
1.1 Stakeholder Identification	20	\$25	\$500		
1.2 Project Charter	30	\$75	\$750		
1.3 Business Case Development	50	\$25	\$1,250		
2. Planning				\$3,000	2%
2.1 Team Planning Meeting	8	\$75	\$700		
2.2 Team Contracts	8	\$75	\$1,500		
2.3 Scope Statement	47	\$75	\$750		
2.4 Work Breakdown Structure (WBS)	16	\$25	\$90		
2.5 Schedule and Cost Estimate				\$700	
2.5.1 Task Resources	4	\$75	\$100		
2.5.2 Task Durations	4	\$75	\$100		
2.5.3 Task Dependencies	9	\$25	\$300		
2.5.4 Task Prioritization	17	\$75	\$300		
2.5.5 Stakeholder Management Strategy	17	\$75	\$90		
3. Executing				\$128,968	79%
3.1 Market Research & User Surveys	20	\$30	\$600		
3.2 AI Algorithm Development				\$128,368	
3.2.1 Smart Fridge Hardware & Sensors	1	\$12,000	\$12,000		
3.2.2 IoT Devices & Modules	1	\$8,000	\$8,000		
3.2.3 AI Development	1	\$29,400	\$29,400		
3.2.4 IoT Software & Integration	1	\$9,000	\$9,000		

Testing (10% of total development costs)				\$11,897	\$11,897	6%
Training & Support					\$2,200	1%
5.1 User Training	10	\$20	\$200			
5.2 Documentation			\$1,000			
5.3 Deployment Support			\$1,000			
Reserves (20% of total estimate)				\$30,913	\$30,913	17%
Digital Innovation Forum					\$6,000	3%
7.1 Display Booth Design & Setup				\$2,250		
7.1.1 Booth Rental & Space Fee	1	\$600	\$600			
7.1.2 Booth Design & Branding Materials	1	\$400	\$400			
7.1.3 Display Screens & AV Equipment	1	\$250	\$250			
7.1.4 Smart Fridge Physical Display Unit	1	\$700	\$700			
7.1.5 Signage & Printed Materials	1	\$300	\$300			
7.2 Prototype Presentation & Testing				\$1,500		
7.2.1 Prototype Assembly & Transport	1	\$500	\$500			
7.2.2 Demo Software Setup & Configuration	4	\$50	\$200			
7.2.3 Technical Support Staff (2 days x 8 hrs)	10	\$50	\$500			
7.2.4 Feedback Collection Tools (Tablets/Surveys)	2	\$150	\$300			
7.3 Investor & Customer Engagement				\$1,000		
7.3.1 Promotional Giveaways & Branded Items	1	\$400	\$400			
7.3.2 Marketing Collateral (Brochures, Flyers)	1	\$300	\$300			
7.3.3 Business Card Printing	100	\$3	\$300			
7.4 Logistics & Operations				\$1,250		
7.4.1 Team Travel & Accommodation (3 people)	1	\$500	\$500			
7.4.2 Catering & Refreshments	1	\$500	\$500			
7.4.3 Insurance & Miscellaneous	1	\$250	\$250			
Total project cost estimate					\$185,478	

* See software development estimate

SmartFridge Software Development Estimate

1. Labor Estimate	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	Calculations
AI Developer labor estimate	400	60	24,000	700 x 60
IoT Developer labor estimate	320	50	16,000	560 x 50
Total labor estimate			40,000	Sum above two values
2. Function point estimate	Quantity	Factor	Points	Calculations
External inputs	9	4	36	10 x 4
External interface files	3	7	21	4 x 7
External outputs	5	5	25	5 x 5
External queries	1	4	4	4 x 4
Logical internal tables	7	10	70	7 x 10
Total function points			156	Sum above function point values
Java 2 language equivalency value			46	Assumed value from reference
Source lines of code (SLOC) estimate			7,176	156 x 46
Productivity x KSI/OC*Penalty (in months)	25.89			3.13 x 0.05*1.072 (see reference)
Total labor hours (27 hours/function point)*	4,212			156 x 27
Cost/labor hour	14			\$AR/hour (assumed from budget expert)
Total function point estimate			58,968	4212 x 14
* Software Development Estimate (MAX of Labor vs FP)			58,968	Take the larger estimate

* Based on historical data

The initial predicted cost was SAR 140,000. However, after conducting a detailed cost estimate aligned with the Work Breakdown Structure (WBS), the total project cost increased to approximately SAR 185,478. This variance of SAR 45,478 represents a funding gap that requires corrective action. To address this, the project team will pursue one or more of the following strategies: requesting increased investment from existing sponsors, Nada Al-Harthi and Al Rajhi Bank, or seeking additional financing through a commercial.

Work Breakdown Structure:

1.0 Initiating

- 1.1 Stakeholder identification & management strategy
- 1.2 Project charter
- 1.3 Business case development
- 1.4 Kick-off meeting

2.0 Planning

- 2.1 Team planning meeting
- 2.2 Team contract
- 2.3 Scope statement
- 2.4 Work Breakdown Structure (WBS)
- 2.5 Schedule and cost baseline
 - 2.5.1 Task resources
 - 2.5.2 Task durations
 - 2.5.3 Task dependencies
- 2.6 Risk prioritization

3.0 Executing

- 3.1 Market research & user surveys
- 3.2 AI algorithm development
 - 3.2.1 Food expiration tracking module
 - 3.2.2 Recipe recommendation engine
- 3.3 IoT sensor integration
 - 3.3.1 Hardware selection & procurement
 - 3.3.2 Sensor-fridge integration
 - 3.3.3 Data pipeline setup

- 3.4 Smart fridge interface (built-in screen)
- 3.5 System security & data protection
- 3.6 System testing & quality assurance
- 3.7 Pilot launch (50 users)
- 3.8 Project benefits measurement

4.0 Monitoring and Controlling

- 4.1 Progress reports (weekly)
- 4.2 Budget tracking
- 4.3 Change requests
- 4.4 Stakeholder status meetings

5.0 Closing

- 5.1 Final project report
- 5.2 Final project presentation
- 5.3 Sponsor sign-off
- 5.4 Lessons learned

Milestone List :

MILESTONE	DATE	STATUS	RESPONSIBLE	ISSUES / COMMENTS
Initiating				
Stakeholder identification & management strategy completed	April 3, 2026	Completed	Rahaf & Team	Stakeholder register prepared
Business case completed	April 6, 2026	Completed	Rahaf	NPV: SAR 272,800 / ROI: 112%
Project charter signed	April 10, 2026	Completed	Rahaf	
Project kick-off meeting held	April 15, 2026	Completed	Rahaf	Went very well
Planning				
Team contract signed	April 15, 2026	Completed	Rahaf	
Scope statement completed	April 25, 2026	Completed	Rahaf	
WBS completed	May 1, 2026	Completed	Rahaf	
Risk prioritization completed	May 10, 2026	Pending	Jumanh	Reviewed with sponsor
Schedule and cost baseline completed	May 20, 2026	Pending	Deem	
Executing				
Market research & user surveys completed	June 1, 2026	Pending	Jumanh	
AI algorithm – expiration tracking completed	July 15, 2026	Pending	Aljazi	

AI algorithm – recipe engine completed	August 1, 2026	Pending	Aljazi
IoT sensor integration completed	August 15, 2026	Pending	Ghada
Smart fridge interface (built-in screen) completed	September 15, 2026	Pending	Ghada
System security & data protection implemented	October 1, 2026	Pending	Aljazi
System testing & QA completed	November 1, 2026	Pending	Ghada
Pilot launch (50 users) completed	December 15, 2026	Pending	Rahaf
Project benefits measurement completed	February 1, 2027	Pending	Deem

Monitoring and Controlling

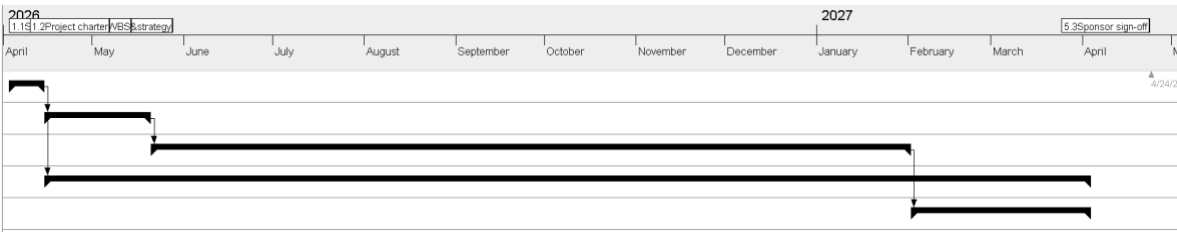
Progress reports	Every Saturday	Ongoing	All	Weekly via Microsoft Teams
------------------	----------------	---------	-----	----------------------------

Closing

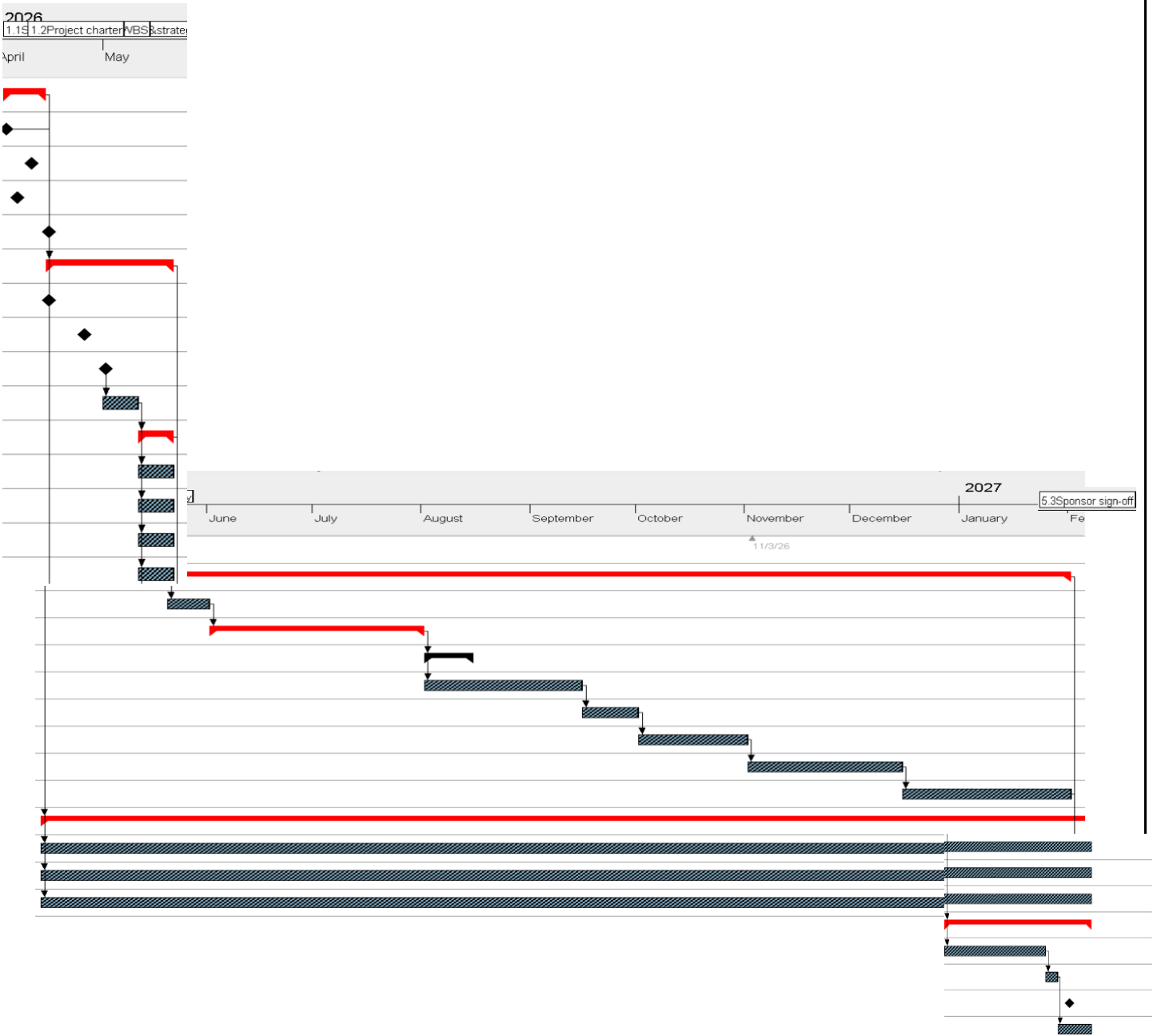
Final project report completed	March 15, 2027	Pending	Rahaf
Final project presentation completed	March 20, 2027	Pending	Rahaf
Sponsor sign-off on project completed	March 25, 2027	Pending	Nada Al-Harhi
Lessons-learned report submitted	April 3, 2027	Pending	All

Gantt chart

General overview

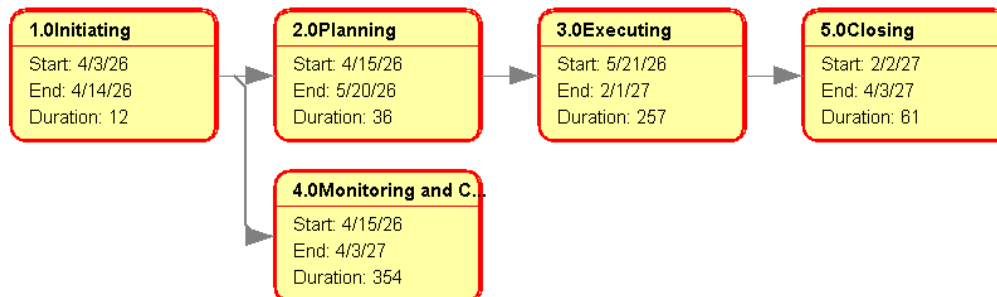


Details

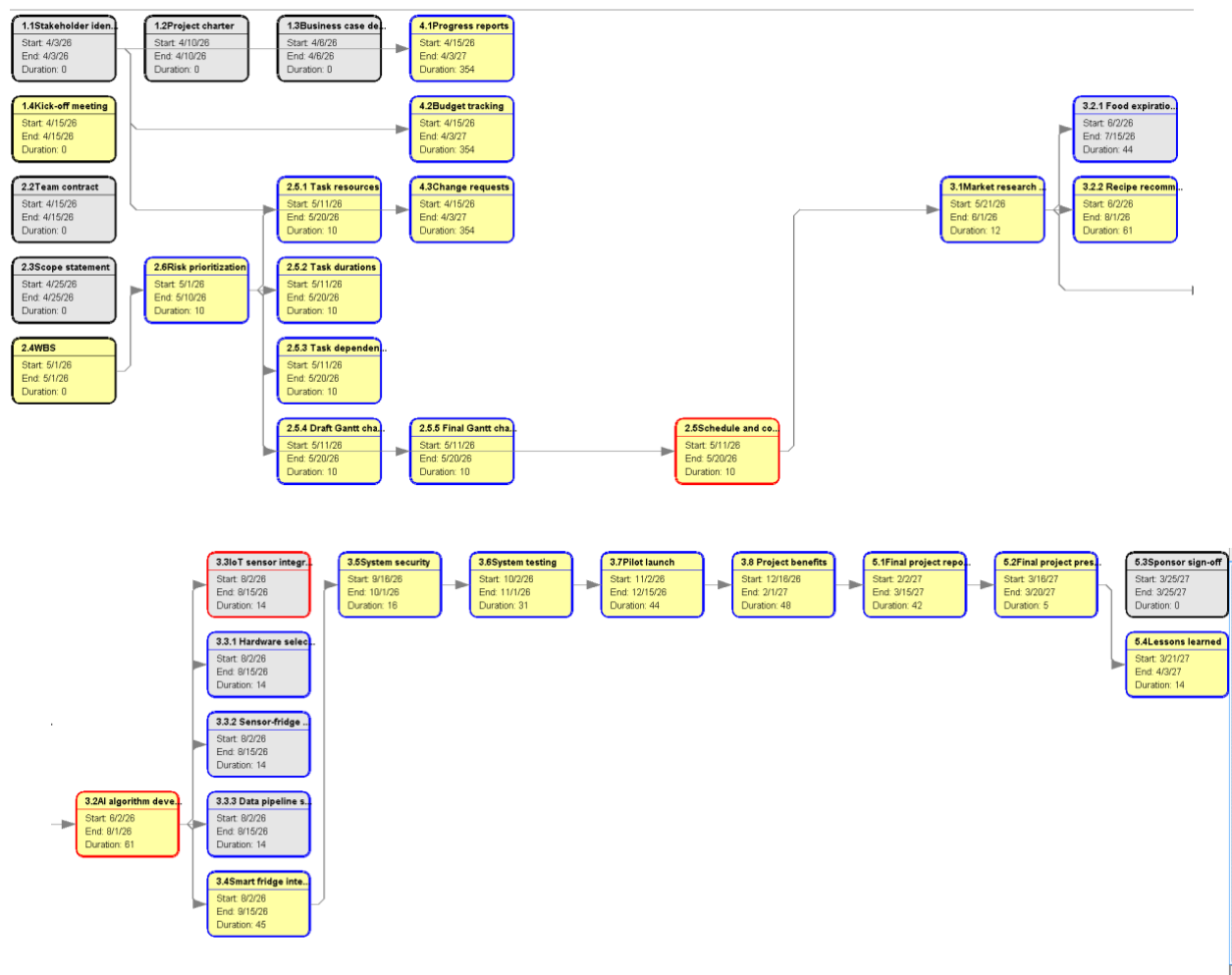


Network diagram

General overview



Details



Business Plan

Executive Summary

SmartFridge is an AI-powered smart refrigerator system designed to help households and restaurants reduce food waste and manage food more efficiently. The system uses Artificial Intelligence (AI) and Internet of Things (IoT) technologies to track food expiration dates, send alerts before products expire, and suggest recipes based on available ingredients.

Saudi Arabia faces high levels of food waste, while most existing smart refrigerators focus mainly on connectivity and remote monitoring features rather than intelligent food management. SmartFridge addresses this issue by providing a smart system that improves food tracking and helps users reduce unnecessary waste.

The project supports Saudi Vision 2030 goals related to digital transformation and sustainability. The estimated project cost is SAR 185,478, with projected annual revenue of SAR 200,000.

Company Description

SmartFridge is a technology-focused project that develops smart food management solutions using AI and IoT technologies. The system helps users monitor food freshness, organize inventory, and improve food usage efficiency.

The project team includes members specialized in Management Information Systems, Artificial Intelligence, Electrical Engineering, and Financial Management. The team works together to build an innovative system that supports smart home technologies and reduces food waste.

Products and Services

SmartFridge provides a smart food management system that helps users organize and monitor food items inside the refrigerator.

- Food Expiration Tracking:

The system automatically tracks food expiration dates using AI algorithms and IoT sensors to help users avoid forgetting food items.

- Smart Alerts:

Users receive automated notifications before food products expire, helping reduce food waste and improve food usage.

- Recipe Recommendations:

The system suggests recipes based on the ingredients currently available inside the refrigerator.

- Inventory Dashboard:

Users can monitor food inventory, quantities, and expiration dates through a simple mobile application and web interface.

- Smart Refrigerator Integration:

SmartFridge integrates with IoT-connected refrigerators to provide real-time monitoring and automatic updates.

Marketing Plan

The SmartFridge marketing strategy focuses on increasing awareness and attracting customers through digital and physical channels.

- Awareness Phase:

Social media platforms such as Instagram, Snapchat, TikTok, and X will be used to promote the system and explain its features.

- Customer Acquisition:

The project will participate in technology exhibitions and product demonstrations in malls and public events. Partnerships with smart appliance stores will also help increase market reach.

- Retention and Growth:

SmartFridge will continue improving user experience and expanding partnerships with restaurants and hospitality businesses.

Operational Plan

- Development Process:

The system will be developed using an iterative development approach across 12 months, including planning, development, testing, and launch phases.

- User Experience:

The system is designed with a simple and user-friendly interface to ensure easy interaction for all users.

- Data Security and Privacy:

SmartFridge will apply security measures to protect user information and maintain data privacy.

- Customer Support:

Customer support services will be available through the application and website to assist users with technical issues and inquiries.

- Scalability and Infrastructure:

The system infrastructure will support future expansion and increasing numbers of users.

Revenue Model

SmartFridge plans to generate revenue through several sources:

- Product Sales:

Selling SmartFridge hardware and software systems.

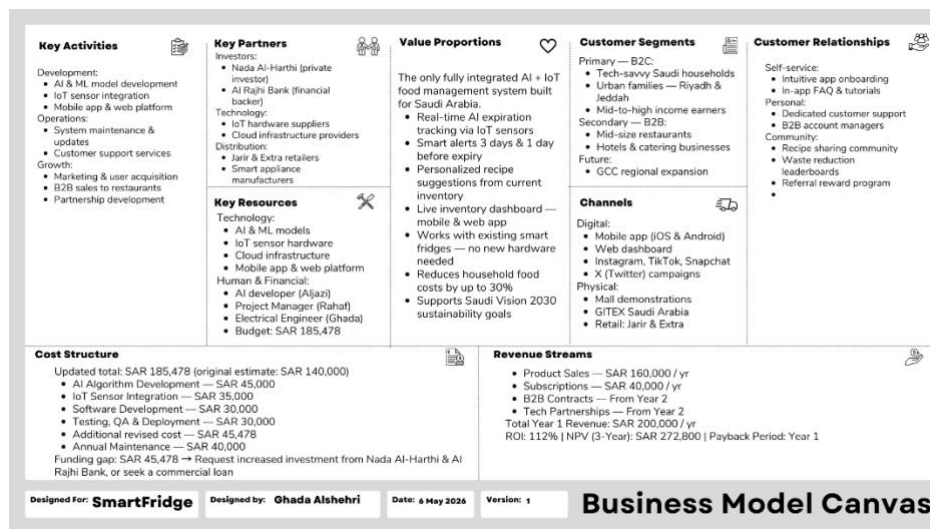
- Partnerships:

Collaborating with smart appliance companies and technology providers.

- Annual Revenue:

The project is expected to generate approximately SAR 200,000 annually.

Business Model



probability/impact matrix

Project Name: SmartFridge

Prepared by: Deem Althawab

Date: May 7, 2026

Impact \ Probability	Low Probability	Medium Probability	High Probability
High Impact	Legal or regulatory issues	Budget overrun (exceeding SAR 140,000)	Failure to meet project deadline
Medium Impact	Low user adoption after launch	Technical integration issues (AI & IoT)	Data security and privacy breaches
Low Impact	Minor hardware delays	Inadequate testing before deployment	Communication gaps between teams

List of Prioritized Risks:

Prepared by: Deem Althawab , Date: May 4th ,2026

Ranking	Potential Risks
1	Failure to meet the project deadline
2	Budget overrun (exceeding SAR 140,000)
3	Failure to satisfy stakeholder requirements
4	Technical issues with AI or IoT integration
5	Data security and privacy breaches
6	Low user adoption after launch
7	Unforeseen legal or regulatory issues
8	Security vulnerabilities in the system
9	Inadequate testing before deployment
10	Integration delays between hardware and software

Updated List of Prioritized Risks

Prepared by: Deem Althawab , Date: May 4th ,2026

Ranking	Potential Risks	Priority Level
1	Failure to meet the project deadline	High
2	Budget overrun	High
3	Failure to satisfy stakeholder requirements	High
4	Technical integration issues	Medium
5	Failure to satisfy stakeholder requirements	Medium
6	Low user adoption	Medium
7	Legal and regulatory issues	Medium
8	Inadequate testing before deployment	Medium
9	Communication gaps between teams	Low
10	Hardware and software integration delays	Low

Risk Register

prepared by:	Deem Althawab	Date: May 5 th , 2026
Rank:	1	6
Risk	Failure to meet the deadline	Data Security and Privacy Breaches
Description:	Delays in development phases may result in missing the planned project completion date.	Unauthorized access to sensitive user data may lead to financial loss, legal consequences, and reputational damage.
Category:	Schedule Risk	Cybersecurity Risk
Root Causes	Underestimation of development time	Weak security infrastructure Lack of proper encryption

	Complex AI and IoT integration Ineffective time planning	Insufficient access control-
Triggers:	Delay in task completion Increased workload beyond plan	Suspicious login attempts System vulnerabilities identified during testing
Potential Response:	Use milestone tracking Conduct weekly progress meetings	Apply strong authentication mechanisms Conduct regular security testing
Risk Owner:	Project Manager	Project Manager
Probability:	High	High
Impact	High	High
Status:	Open	Open

Lessons Learned Report:

Project Name: SmartFridge

Project Manager: Rahaf Al-Harbi

Project Duration: April 2026 – April 2027

Project budget: SAR 185,478

1. Did the project meet scope, time, and cost goals?

The SmartFridge project is currently on track to meet its planned scope, schedule, and cost objectives. The project scope is being delivered as defined, while time and budget are being closely monitored through regular progress tracking and risk management. Some minor delays were identified, but they are being controlled and managed effectively.

2. What was the success criteria listed in the project scope statement?

The success criteria of the SmartFridge project included:

- Delivering all required features such as smart monitoring, IoT connectivity, and AI-based recommendations.
- Completing the project within the scheduled timeline (April 2026 – April 2027).
- Staying within the allocated budget of SAR140,000.
- Ensuring system reliability, security, and user-friendly design.
- Achieving positive user acceptance and satisfaction after deployment.

3. Reflect on whether or not you met the project success criteria.

The project is currently progressing toward meeting all success criteria. The functional requirements are being developed according to the scope, and quality assurance processes are helping maintain system reliability. While some technical challenges (especially in AI and IoT integration) caused slight delays, the team is actively resolving them. Overall, the project is expected to meet its success criteria if the current management approach continues.

4. In terms of managing the project, what were the main lessons your team learned?

The main lessons learned include:

- Early planning is essential for avoiding delays and misestimating technical complexity.
- Continuous communication between team members and stakeholders improves project alignment.
- Risk management should be applied from the beginning, not only during problems.
- Regular testing is critical for identifying integration issues early.
- Clear role assignment improves team efficiency and reduces confusion.

5. Describe one example of what went right on this project.

One major success was the strong collaboration between team members. Tasks were clearly divided, and communication remained consistent throughout the project. This helped the team maintain steady progress and quickly resolve issues when they appeared.

6. Describe one example of what went wrong on this project.

One issue that occurred was underestimating the technical complexity of integrating AI and IoT systems. This led to unexpected delays in the early development phase and required additional time for testing and adjustments.

7. What will you do differently on the next project based on your experience working on this project?

On future projects, the team will:

- Improve time and cost estimation accuracy during the planning phase.
- Conduct deeper technical feasibility analysis before development starts.
- Involve stakeholders earlier to clarify requirements.
- Allocate more time for testing and integration phases.
- Strengthen risk management planning from the beginning of the project lifecycle.
- Enhance documentation and tracking systems to improve project visibility

Best Practices:

1. Define clear project objectives and requirements from the beginning.
2. Maintain regular communication with stakeholders and team members.
3. Monitor project progress continuously using milestones and reports.
4. Assign clear roles and responsibilities to all team members.
5. Conduct regular testing and evaluation during development.
6. Identify and manage risks early to reduce delays and unexpected issues.
7. Keep proper documentation for all project activities and updates.
8. Encourage teamwork and knowledge sharing among the project team.
9. Focus on user feedback to improve usability and performance.
10. Continuously learn from challenges and apply improvements throughout the project lifecycle.

conclusion

SmartFridge is an innovative AI-powered smart kitchen solution that combines Artificial Intelligence and Internet of Things technologies to improve food management and reduce food waste.

The project supports Saudi Vision 2030 goals related to sustainability and smart technology adoption. By providing automated food tracking, smart alerts, and recipe recommendations, SmartFridge aims to help households and restaurants improve food management efficiency and reduce unnecessary waste.

References:

- Schwalbe, K. (2022). Information Technology project management (9th ed.). Cengage Learning.
- Russell, S., & Norvig, P. (2021). Artificial intelligence: A modern approach (4th ed.). Pearson.
- Gubbi, J., Buyya, R., Marusic, S., & Palaniswami, M. (2013). Internet of Things (IoT): A vision, architectural elements, and future directions. *Future Generation Computer Systems*, 29(7), 1645–1660.
<https://doi.org/10.1016/j.future.2013.01.010>
- Food and Agriculture Organization. (2019). The state of food and agriculture: Moving forward on food loss and waste reduction. FAO. <https://www.fao.org/publications/sofa/2019/en/>
- Saudi Vision 2030. (2016). Vision 2030 Kingdom of Saudi Arabia. <https://www.vision2030.gov.sa>
- Statista. (2024). Kitchen appliances market revenue in Saudi Arabia.
<https://www.statista.com/outlook/cmo/household-appliances/saudi-arabia>
- Osterwalder, A., & Pigneur, Y. (2010). Business model generation: A handbook for visionaries, game changers, and challengers. John Wiley & Sons.